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Design Document

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1 Introduction

1.1 Purpose

BBP is a software designed to simplify the life of everyone who goes on bike trips and wants to enjoy them without worrying about finding the best routes.

It is intended as a community for cyclists, in which every person can share information about their personal routes, so that everyone may take advantage of the information and experience of other bikers. The application is intended for smartphone, where it can make use of GPS and sensor data to seamlessly reconstruct the path a cyclist has taken.

1.2 Scope

The scope of this Design Document (DD) is to define the technical architecture and the detailed design of the Best Bike Paths (BBP) system.

It translates the requirements specified in the RASD into a modular, microservice-based structure, detailing the components, their interfaces, and the user interface that represents the implementation of the system.

It represents the structure to respect in development to ensure that the system is scalable, maintainable, and that it correctly provides a reliable experience for users.

1.3 Definitions, Acronyms, Abbreviations

Acronym	Definition
BBP	Best Bike Paths application
GPS	Global Positioning System
UI	User Interface
SP	Shared Phenomenon
WP	World Phenomenon
UC	Use Case
DA	Domain Assumption
G	Goal

Table 1: Definitions, Acronyms and Abbreviations

1.4 Reference Documents

- Requirements analysis and specification document (RASD) of Best Bike Paths application
- Lecture slides and notes from the course Software Engineering 2 2025/2026

1.5 Document Structure

This document is organized into five main sections, each covering a specific aspect of the BBP (Best Bike Paths) system design:

- **Section 1 - Introduction:** This section provides an overview of the document, defining its purpose and scope. It also includes a list of definitions, acronyms, and abbreviations used throughout the text, along with references to related documents.
- **Section 2 - Architectural Design:** This section describes the system's high-level architecture. It details the microservice-based approach through a Component View and a Deployment View.

Furthermore, it includes a Runtime View where sequence diagrams illustrate how various use cases are executed across the components, and a dedicated subsection for Component Interfaces.

- **Section 3 - User Interface Design:** This section presents the design of the mobile application's interface. It explains the layout and navigation flow, highlighting how the UI has been structured to satisfy the specific needs and goals of the end-users.
- **Section 4 - Requirements Traceability:** This section provides a mapping between the architectural components defined in this document and the functional requirements previously identified in the RASD (Requirements Analysis and Specification Document), ensuring every requirement is addressed by the design.
- **Section 5 - Implementation, Integration, and Test Plan:** The final section outlines the development strategy. It identifies the order in which subcomponents will be implemented and describes the plan for integrating and testing them to ensure system reliability.
- **Section 6 - Effort Spent:** This section provides an overview of the effort spent by each team member during the creation of this document.

2 Architectural Design

3 User Interface Design

In this section, we describe the user interface of our application. The interface is designed to be intuitive and user-friendly, allowing users to easily navigate through the various features and functionalities of BBP.

Given the nature of BBP, the user interface is user-friendly, providing easy access to all the features and functionalities.

Pages and Navigation

The application is structured into several pages to ensure a logical user flow and an intuitive experience.

The following pages represent the core architecture of the platform:

- **Login and Register Page:** Handles user authentication and registering, allowing users to securely access their accounts or create new ones.
- **Home Page:** Serves as the primary dashboard for users, for inserting and starting registering trips and visualizing recent activities.
- **Search Page:** Enables users to search for published routes on the map with all information about the trip.
- **Your Trips Page:** A central hub for managing personal travel plans, displaying both upcoming itineraries and historical booking data.
- **Account Page:** Dedicated to user profile management, where individuals can update personal information, adjust security settings, and manage application preferences.

Apart from the Login and Register Page, that is the first page you encounter when opening the application the first time, after the login all other pages are accessible via a bottom navigation bar, allowing users to switch between them easily.

3.1 Login and Register Page

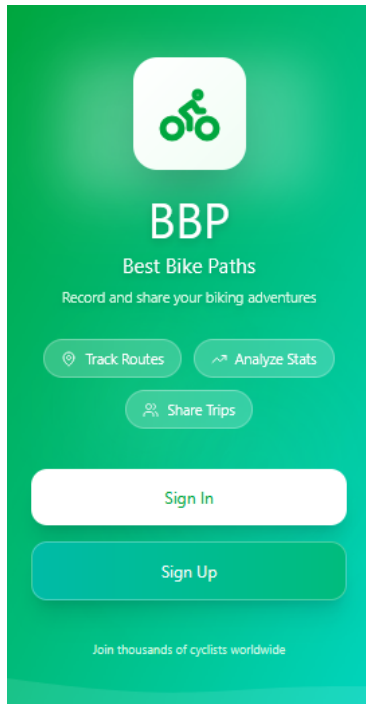


Figure 1: Initial Page

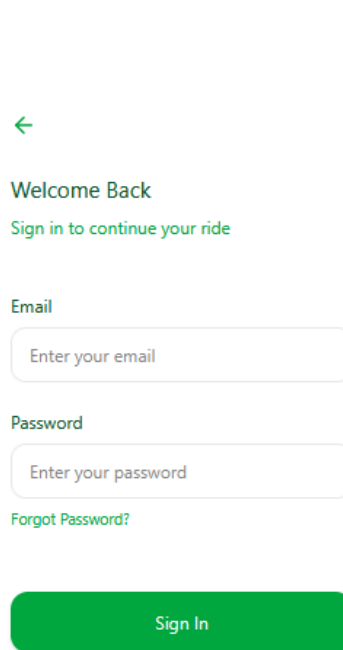


Figure 2: Login Page

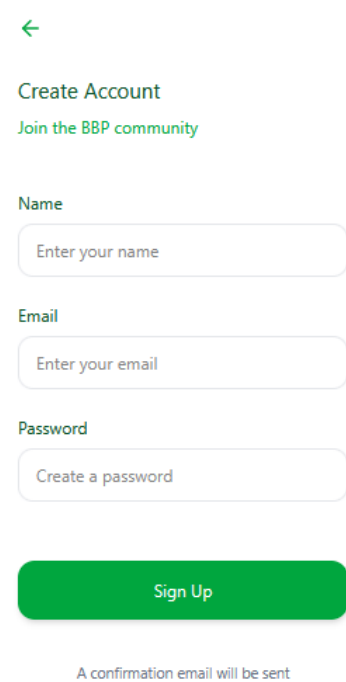


Figure 3: Register Page

The first page when you're not logged in is shown in Figure 1, where you can choose to log in or register a new account.

If you choose to log in, you will be directed to the Login Page shown in Figure 2, where you can enter your credentials to access your account.

If you don't have an account yet, you can click on the "Sign Up" button to be directed to the Register Page shown in Figure 3, where you can create a new account by providing the required information.

3.2 Home Page

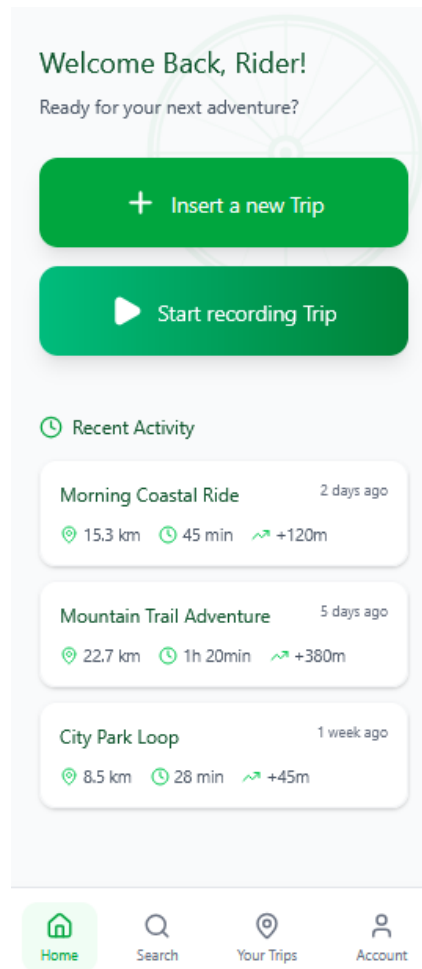


Figure 4: Home Page

The Home Page, shown in Figure 4, serves as the main dashboard for users.

Here, users can manually insert a trip by clicking on the "Insert a new Trip" button, that will open a form that user have to compile to submit the trip, or start the automatic recording of a new trip by clicking on the "Start recording Trip" button, which will guide them through the trip registration process.

Additionally, the Home Page displays recent activities and updates related to the user's account.

3.3 Search Page

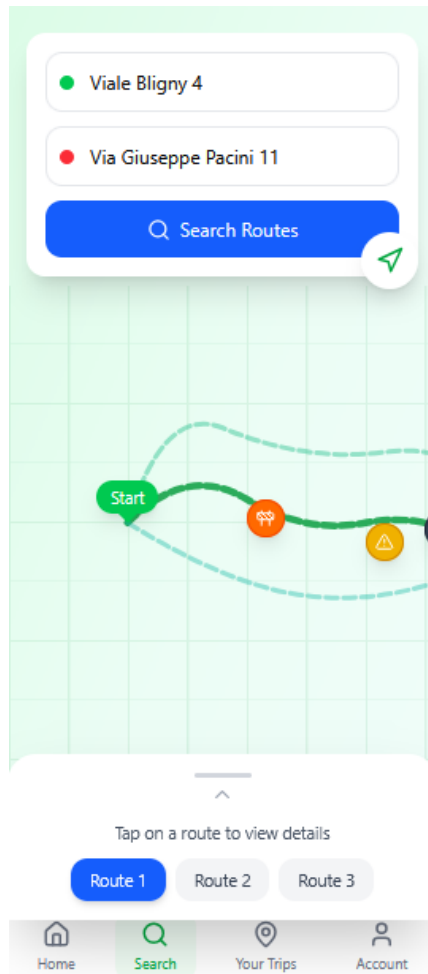


Figure 5: Search Page

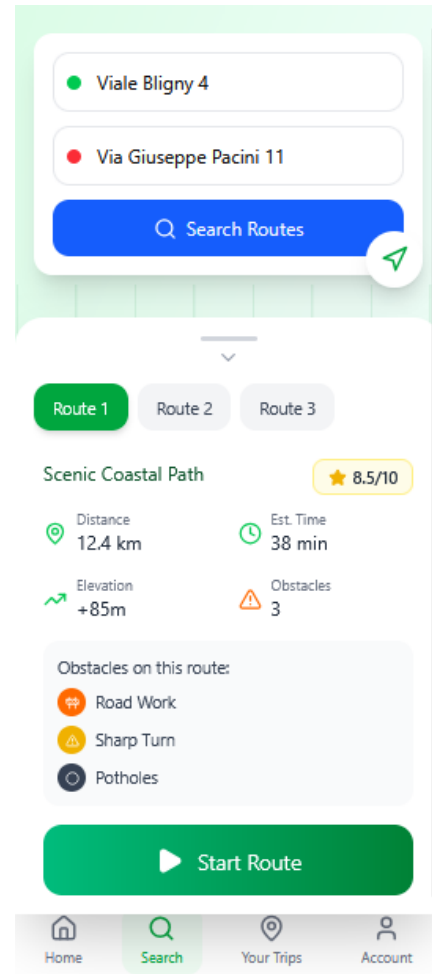


Figure 6: Route Details

The Search Page, shown in Figure 5, allows users to search for published routes on the map inserting start and end point in the forms on the upper part of the page and clicking on them on the map to view detailed information about each trip, as illustrated in Figure 6.

3.4 Your Trips Page

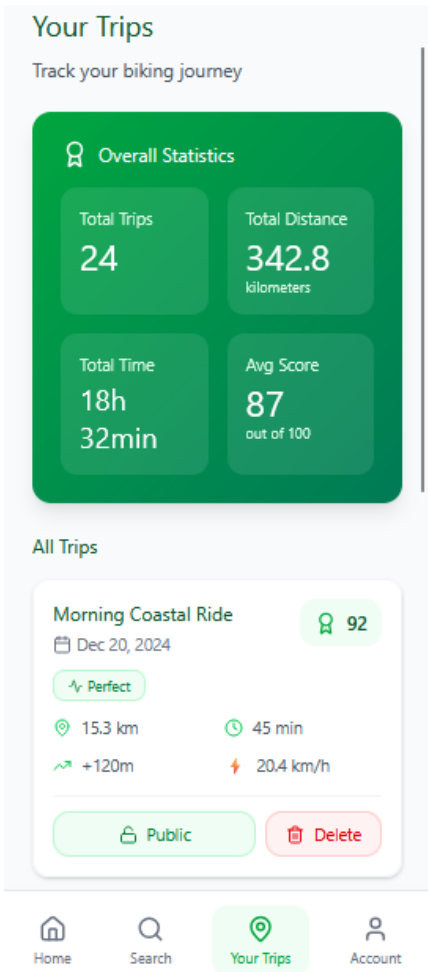


Figure 7: Your Trips Page

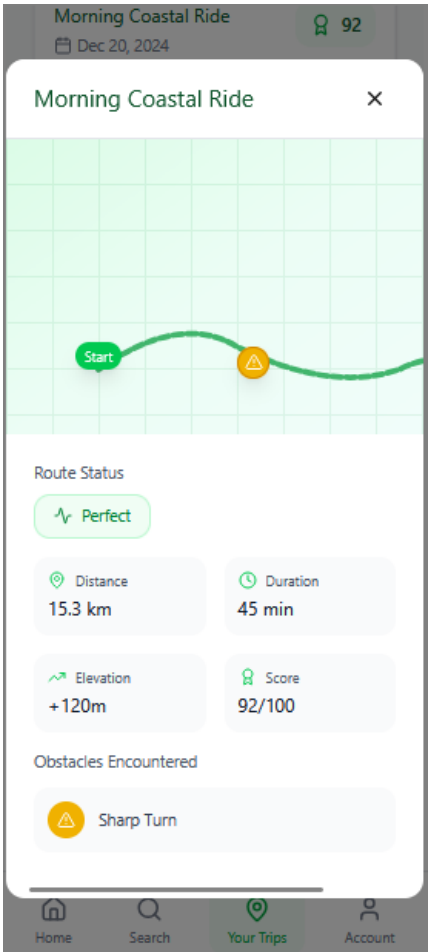


Figure 8: Trip Details

The Your Trips Page, shown in Figure 7, is where users can manage their personal travel plans. It displays both upcoming itineraries and historical booking data. By clicking on a specific trip, users can view detailed information about that trip, as shown in Figure 8.

3.5 Account Page

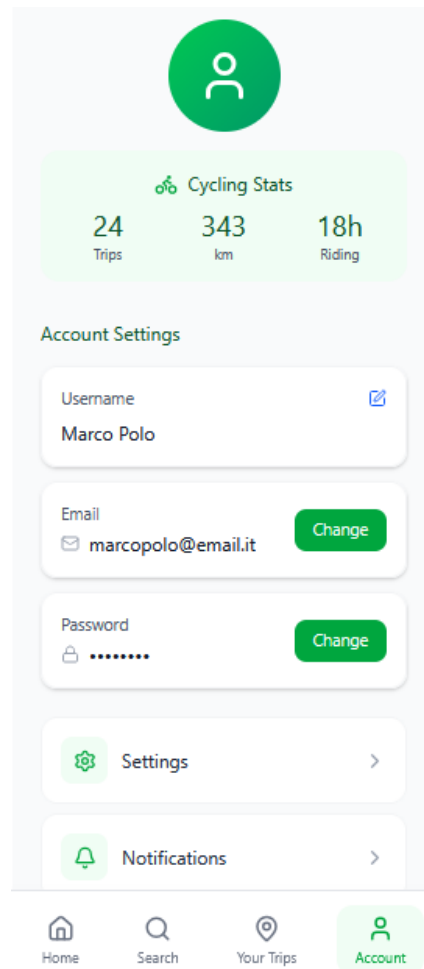


Figure 9: Account Page

The Account Page, shown in Figure 9, is dedicated to user profile management.

Here, users can update their personal information like name, email, and password and manage application preferences according to their needs.

4 Requirements Traceability

The following table lists the traceability between the identified requirements and the corresponding components and modules implemented in the system.

Table 2: Requirements traceability

Component	Requirements
Mobile Application	R1: System accepts account registration data R5: System allows users to login to their account by authenticating via email (or username) and password R6: System allows users to logout of their account R7: System allows users to modify their account information (username and personal data) R8: System allows users to modify/recover their password via email verification R9: System allows users to delete their account, with all the associated data R10: System allows users to start automated trip recording R11: System requests necessary sensor and GPS permissions when automated recording is started R14: System collects GPS coordinates during automated trip recording R15: System reconstructs traveled path from collected GPS coordinates R16: System collects data from mobile device's accelerometer and gyroscope during automated trip recording R18: System allows users to manually stop automated trip recording R19: System allows users to manually record trips by inserting starting and ending locations, names of roads in traversal order, start and end times R20: System allows users to specify road status (optimal, medium, sufficient, requires maintenance) during manual recording R21: System allows users to specify obstacle locations during manual recording R32: System displays a list of user's recorded trips showing name, date, start and end locations R33: System allows users to view detailed trip data for any recorded trip, including route map, statistics and weather data R34: System allows users to manually edit trip data for any of their recorded trips R35: System allows users to delete any of their recorded trips R37: System allows users to make published trips private again R38: System allows registered users to publish their recorded trips, making them publicly available R39: System requires the user to correct and confirm data before publishing, if it was acquired automatically R40: System allows users to add/remove obstacle markers during data correction R41: System allows users to modify road status during data correction R42: System allows all users to view published bike paths between specified start and end points R46: System displays all identified paths on a map, with the highest scoring path highlighted and the others with reduced opacity R49: System allows users to select a displayed path R51: System allows users to export selected path to compatible third-party navigation applications

Table 2: Requirements traceability

Component	Requirements
API Gateway	R1: System accepts account registration data R5: System allows users to login to their account R6: System allows users to logout of their account R7: System allows users to modify their account information R8: System allows users to modify/recover their password R9: System allows users to delete their account, with all the associated data R32: System displays a list of user's recorded trips R33: System allows users to view detailed trip data R34: System allows users to manually edit trip data R35: System allows users to delete any of their recorded trips R37: System allows users to make published trips private again R38: System allows registered users to publish their recorded trips R42: System allows all users to view published bike paths R46: System displays all identified paths on a map R49: System allows users to select a displayed path R51: System allows users to export selected path to compatible third-party applications
Service Discovery	Enables communication between components for all requirements
User Service	R1: System accepts account registration data R2: System rejects account registrations with duplicate emails or usernames R3: System sends a verification email with an OTP to the email address provided during registration R4: System must verify OTP code before confirming registration R5: System allows users to login to their account by authenticating via email (or username) and password R6: System allows users to logout of their account R7: System allows users to modify their account information (username and personal data) R8: System allows users to modify/recover their password via email verification R9: System allows users to delete their account, with all the associated data

Table 2: Requirements traceability

Component	Requirements
Trip Service	R12: System begins automated path and sensor tracking, after recording initiation, only when user has started biking R19: System allows users to manually record trips by inserting starting and ending locations, names of roads in traversal order, start and end times R22: System persistently stores recorded trips R23: System ensures only author of recorded trips can access unpublished trip records R25: System calculates statistics immediately after trip is saved R32: System displays a list of user's recorded trips showing name, date, start and end locations R33: System allows users to view detailed trip data for any recorded trip, including route map, statistics and weather data R34: System allows users to manually edit trip data for any of their recorded trips R35: System allows users to delete any of their recorded trips R36: System propagates modifications and deletions of trips to their published versions, if present R37: System allows users to make published trips private again R38: System allows registered users to publish their recorded trips, making them publicly available
Data Processing Service	R13: System algorithmically detects if a user has started biking from speed data R15: System reconstructs traveled path from collected GPS coordinates R17: System algorithmically interprets collected sensor data to detect potential obstacles along the path R24: System automatically calculates statistics for each recorded trip (distance, average speed, duration, elevation, average slope) R26: System calculates aggregate statistics across all of user's recorded trips R27: System recomputes aggregate statistics upon modification of user's trip record (add/delete/edit trip) R29: System infers road status from detected obstacle density and movement pattern analysis when trip data is recorded automatically R30: System queries an external meteorological service for weather data to enrich trip data after completion

Table 2: Requirements traceability

Component	Requirements
Path Service	R38: System allows registered users to publish their recorded trips, making them publicly available R42: System allows all users to view published bike paths between specified start and end points R43: System identifies all published bike paths that connect specified start and end points R44: System notifies user when no published paths exist between the specified points R45: System calculates a score for each identified bike path based on road status, route effectiveness (distance and directness in reaching destination), amount of obstacles, path difficulty (slope) R47: System displays score values for each path R48: System initially displays a limited amount of the identified paths, providing the option to view all paths R49: System allows users to select a displayed path R50: System allows users to view additional information about selected path (status, obstacles)
Message Broker	R24: System automatically calculates statistics for each recorded trip (distance, average speed, duration, elevation, average slope) R30: System queries an external meteorological service for weather data to enrich trip data after completion R36: System propagates modifications and deletions of trips to their published versions, if present R38: System allows registered users to publish their recorded trips, making them publicly available
Databases & DBMS	R2: System rejects account registrations with duplicate emails or usernames R9: System allows users to delete their account, with all the associated data R22: System persistently stores recorded trips R23: System ensures only author of recorded trips can access unpublished trip records R28: System persistently stores computed statistics with corresponding trip data R31: System stores a notice in trip data when weather data is unavailable R35: System allows users to delete any of their recorded trips R36: System propagates modifications and deletions of trips to their published versions, if present R43: System identifies all published bike paths that connect specified start and end points
Map Provider	R33: System allows users to view detailed trip data for any recorded trip, including route map, statistics and weather data R42: System allows all users to view published bike paths between specified start and end points R46: System displays all identified paths on a map, with the highest scoring path highlighted and the others with reduced opacity R48: System initially displays a limited amount of the identified paths, providing the option to view all paths

Table 2: Requirements traceability

Component	Requirements
Weather Provider	R30: System queries an external meteorological service for weather data to enrich trip data after completion R33: System allows users to view detailed trip data for any recorded trip, including route map, statistics and weather data

5 Implementation, Integration and Test Plan

5.1 Implementation plan

The implementation of BBP will be executed in a phased manner to manage the complexities of a distributed system.

Initially, development will focus on the infrastructure layer using Spring Cloud and Netflix components. This includes setting up the Service Discovery and API Gateway to establish a robust routing mechanism for all incoming client requests.

Once the infrastructure is stable, work will proceed to the User Service to manage registration and authentication processes.

Following this, the Trip Service and Path Service will be developed to handle the core logic of recording cycling activities and computing path scores based on user-submitted status reports.

Finally, the Data Processing Service will be implemented to consume events from the Kafka broker, allowing for the generation of performance statistics and the integration of external meteorological data.

5.2 Integration plan

The integration strategy for BBP combines synchronous and asynchronous patterns to balance performance with reliability. Direct interactions, such as user logins or path searches, will be handled synchronously through the API Gateway via RESTful calls, given the Microservices architecture. This ensures immediate feedback for high-priority user actions.

Instead more complex operations will utilize an event-driven approach. When the Trip Service completes a recording, it will publish a "Trip Recorded" event to the Kafka message broker. This allows the Data Processing Service to aggregate statistics and the Path Service to update the path database without blocking the user's mobile application.

Furthermore, external integrations with Map and Weather providers will be encapsulated within their respective services to maintain the principle of loose coupling of Microservices.

5.3 Test plan

The testing phase ensures that all functional requirements are met and that the system remains stable under load in each development stage and for each microservice within the architecture.

- **Unit Testing:** Each one of the microservices will undergo rigorous unit testing using JUnit to validate business logic in isolation.
- **Integration Testing:** We will verify the exchange of messages through a tool like Kafka and ensure that the three independent databases (User, Trip, and Path) maintain data integrity during each type of operations.
- **Functional and System Testing** This will focus on critical BBP scenarios, such as the accuracy of path score calculations and the reliability of the automated mode in reconstructing GPS paths. We will also simulate concurrent user access to ensure the system can handle the expected community load.
- **User Interface Testing:** The mobile application will be tested on various devices and operating systems to ensure a consistent user experience and responsiveness and ease of use from client perspective.
- **Security Testing:** We will conduct vulnerability assessments and penetration testing to safeguard user data, particularly focusing on authentication and authorization mechanisms, utilizing notorious hacking strategies.

- **Performance Testing:** Load testing will be conducted to assess system performance under peak conditions, ensuring that response time remain acceptable.

6 Effort Spent

Member of group	Effort spent	
Bedini Simone	Introduction	<i>Xh</i>
	Architectural design	<i>Xh</i>
	User interface design	<i>Xh</i>
	Requirements Traceability	<i>Xh</i>
	Implementation, Integration and Test Plan	<i>Xh</i>
	Reasoning	<i>Xh</i>
Cantoli Angelo	Introduction	<i>Xh</i>
	Architectural design	<i>Xh</i>
	User interface design	<i>Xh</i>
	Requirements Traceability	<i>Xh</i>
	Implementation, Integration and Test Plan	<i>Xh</i>
	Reasoning	<i>Xh</i>

7 References

- Requirements analysis and specification document (RASD) of Best Bike Paths application
- Lecture slides and notes from the course Software Engineering 2 2025/2026

7.1 Used Tools

The following tools have been used to create this document:

- **LaTeX**: for document preparation and typesetting.
- **PlantUML**: for creating UML diagrams included in the document.
- **GitHub**: for version control and collaboration on the document source files.
- **Figma**: for designing user interfaces and prototyping.
- **VSCode**: as the main text editor for writing LaTeX code.